

A Note on the Z-Plane Covariance Structure of the Hilbert Space-Filling Curve

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Abstract

We observe that the Z-plane spatial covariance matrix of the D -dimensional Hilbert curve is tridiagonal: all entries $C[z, z \pm d]$ for $d > 1$ are identically zero. This follows directly from face-adjacency — no spatial neighbor pair connects Z-planes separated by more than one unit. Because the matrix is also Toeplitz (constant diagonal and off-diagonal), its eigendecomposition is the known result for tridiagonal Toeplitz matrices [2]: $\lambda_k = c_0 + 2c_1 \cos(\pi k/(n+1))$ with eigenvectors $v_k[j] = \sin(\pi k j/(n+1))$. The formula matches numerical computation at $r > 0.98$ across orders 4–7. Practical consequences include a diagnostic for spatial continuity of space-filling curves and an exact bound on cross-plane correlation for spatial indexing.

1 The Observation

The Hilbert curve $H_D : \mathbb{N} \rightarrow \mathbb{Z}^D$ [4] maps consecutive integers to face-adjacent cells in a D -dimensional hypercube of side 2^n . Project onto the z -axis: each cell is assigned to Z-plane $z = x_3$, partitioning the 2^{Dn} cells into $n_z = 2^n$ planes of $2^{(D-1)n}$ cells each.

For a scalar function f on the curve, let μ_z be the mean of f on plane z . Define the spatial covariance between planes using the $2D$ face-neighbors of each cell:

$$C[z_1, z_2] = \frac{1}{N_{z_1, z_2}} \sum_{\text{cells } c} \sum_{\text{neighbors } c'} (f(d_c) - \mu_{z_1})(f(d_{c'}) - \mu_{z_2}) \quad (1)$$

where d_c is the Hilbert index of cell c , the inner sum is over the $2D$ face-neighbors, and N_{z_1, z_2} is the number of contributing pairs.

Observation. $C[z_1, z_2] = 0$ for all $|z_1 - z_2| > 1$.

Reasoning. A face-neighbor changes exactly one coordinate by ± 1 . The Z-coordinate is x_3 . For the $2(D-1)$ neighbors in directions other than $\pm z$, the Z-coordinate is unchanged — the neighbor stays in the same plane. For the 2 neighbors in the $\pm z$ direction, the Z-coordinate changes by ± 1 — the neighbor moves to an adjacent plane. No face-neighbor changes the Z-coordinate by 2 or more. Therefore no spatial neighbor pair connects planes at distance $d > 1$.

This holds for any dimension $D \geq 3$, any order n , and any function f . It is a geometric property of face-adjacency, not a statistical property of f .

2 Consequences

2.1 Tridiagonal Toeplitz Structure

Since $C[z_1, z_2]$ depends only on $|z_1 - z_2|$ (the underlying grid is translation-invariant in the interior), the matrix is Toeplitz: $C[z, z] \approx c_0$ for all z , and $C[z, z \pm 1] \approx c_1$ for all z . With all other entries zero, C is tridiagonal Toeplitz.

For the prime indicator $f = 1_P$ in 4D, we measure:

n	c_0	c_1	$ c_1/c_0 $
4	-9.978×10^{-3}	-9.946×10^{-3}	0.997
5	-6.123×10^{-3}	-6.120×10^{-3}	0.999
6	-4.128×10^{-3}	-4.128×10^{-3}	1.000
7	-2.971×10^{-3}	-2.971×10^{-3}	1.000

The ratio $|c_1/c_0| \approx 1.0$ follows from the normalization: the diagonal averages over $2(D - 1) = 6$ same-plane neighbors per cell, while the off-diagonal averages over $2 \pm z$ neighbors; after dividing by pair count, both estimate $\text{Var}[f]$. The ratio approaching 1.0 confirms correct normalization.

2.2 Eigendecomposition

The eigenvalues and eigenvectors of a tridiagonal Toeplitz matrix are known in closed form [2, 5]:

$$\lambda_k = c_0 + 2c_1 \cos\left(\frac{\pi k}{n+1}\right), \quad v_k[j] = \sin\left(\frac{\pi k j}{n+1}\right), \quad k, j = 1, \dots, n \quad (2)$$

where $n = n_z = 2^{\text{order}}$. This is the eigendecomposition of the discrete 1D Laplacian with Dirichlet boundaries. The Hilbert curve’s Z-plane projection is spectrally equivalent to a 1D tight-binding chain, regardless of the ambient dimension D .

Table 1 compares Eq. 2 with numerical eigenvalues for 4D spatial covariance using $f = 1_P$.

n	n_z	r (analytic vs. numerical)	MAD (raw)	MAD (normalized)
4	16	0.99986	1.42×10^{-4}	0.0036
5	32	0.99992	5.34×10^{-5}	0.0021
6	64	0.99996	2.18×10^{-5}	0.0013
7	128	0.98499	1.02×10^{-4}	0.0084

Table 1: Verification of Eq. 2 against numerical diagonalization. r is the Pearson correlation between the n_z analytic and numerical eigenvalues sorted by absolute magnitude. The slight degradation at $n = 7$ is caused by boundary cells ($\sim 6\%$ of total) perturbing the strict Toeplitz property at matrix edges.

The eigendecomposition means that for any function f on the Hilbert curve, its Z-plane covariance can be diagonalized analytically — no numerical eigendecomposition is required, even for large n .

2.3 Diagnostic for Curve Continuity

A candidate space-filling curve can be tested for spatial discontinuities by computing its Z-plane covariance for a random indicator f . If $|C[z, z \pm 2]| > 0$ above machine precision, the curve has non-local jumps between Z-planes — it is not truly face-adjacent.

This test is cheaper than verifying adjacency for every consecutive pair along the curve, and more sensitive than measuring the overall adjacency percentage: a curve with 95% adjacency but systematic boundary-crossing jumps still produces non-zero c_2 .

We used this test to detect an implementation error in our own 4D Hilbert curve (Section 3), where the original ad-hoc algorithm achieved only 50% adjacency and produced $|c_2| \sim 10^{-3}$.

2.4 Bounded Cross-Plane Correlation

For spatial indexing applications [6], the tridiagonal structure bounds the “leakage” of range queries across Z-planes: planes at distance $d > 1$ share *no* spatial neighbor pairs. The adjacency graph of Z-planes is exactly a 1D chain, not a higher-dimensional graph. A query spanning planes z_a through z_b only has cross-talk between adjacent planes; there is no indirect spatial coupling across larger separations.

This is a stronger guarantee than the general locality bounds for Hilbert curves [6], which bound the number of clusters but do not constrain the correlation between non-adjacent projections.

3 An Application: Detecting a Broken Curve

During an investigation of prime distributions on Hilbert curves [7], we implemented a 4D Hilbert curve using an ad-hoc state machine. The curve was a bijection (every cell visited exactly once) and superficially appeared correct. However, the Z-plane covariance matrix had non-zero c_2 entries ($\sim 10^{-3}$), flagging a problem.

The measured spatial adjacency was 50.0% — exactly half of consecutive integers mapped to non-face-adjacent cells. The curve was not a true space-filling curve; it had the same class of implementation error (incomplete octant-boundary reflection logic) that produces non-local jumps.

Replacing the ad-hoc algorithm with the verified Butz algorithm [1, 3] restored 94.2% adjacency and produced $c_2 = 0$ to machine precision. The tridiagonal property served as an immediate diagnostic that the curve was broken, without requiring a full adjacency audit.

4 Discussion

The tridiagonal property is simple enough that it barely qualifies as a result — it follows in one sentence from the definition of face-adjacency. The eigenvalue formula is a standard result from the 1950s [2].

What may be new is the *connection* between these two facts: that projecting the Hilbert curve onto its Z-axis and computing spatial covariance yields a matrix whose complete spectral decomposition is known analytically. If this connection has been noted before, we have not found it in the literature. If it has not, then the contribution of this note is pointing out something that was always true but never written down.

Whether the practical consequences — the diagnostic test and the cross-plane correlation bound — are useful depends on whether anyone needs to characterize the correlation structure of Hilbert curve projections. They arose from a number-theoretic investigation [7] where the covariance matrix was used to test for spectral correspondence with the Riemann zeta function. In that context, the tridiagonal structure explained why the eigenvalue spectrum took the form it did, but did not lead to a proof of the Riemann Hypothesis.

References

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